



MOBILE APP “WATCH HUB”

4th SEMESTER

Batch code: PR2-

202401B1+202312B+12b1

Date: January 29, 2026

I would also like to extend my heartfelt appreciation to my project team members for their cooperation, dedication, and teamwork. Their support, idea sharing, and collaborative efforts helped in completing this project smoothly within the given time.

Project Team Members:

Student1525132	MR. AFFAN
Student1505937	MR. HAFIZ MOHSIN ALI
Student1538735	MR.ISMAIL MUNEER
Student1542236	MR.MUHAMMAD UMER
Student1537716	MR.AYAZ KHAN
Student1546648	MR.NABEEL AHMED SIDDIQUI

TABLE OF CONTENTS

ACKNOWLEDGEMENT.....	3
PROBLEM DEFINITION.....	3
INTRODUCTION	4
.....	5
.....	5
.....	6
.....	12
.....	16
.....	15
.....	17
.....	19
.....	20
.....	45
..... :	45
.....	45
.....	45

ACKNOWLEDGEMENT

We would like to express my sincere gratitude to **Almighty Allah** for providing me with the strength and ability to complete this semester project successfully.

We are deeply thankful to our respected instructors and teachers **Sir Mujtaba Ali Baloch** at Aptech Learning Center for their valuable guidance, continuous support, and encouragement throughout the semester. Their knowledge, teaching, and timely feedback played a major role in shaping this project and improving our understanding of the subject.

I would also like to extend my heartfelt appreciation to my project team members for their cooperation, dedication, and teamwork. Their support, idea sharing, and collaborative efforts helped in completing this project smoothly within the given time.

Project Team Members:

Team Member: Affan

Team Member: Muhammad Umer

Team Member: Mohsin Ali

Team Member: Ismail Muneer

Team Member: Ayaz Khan

PROBLEM DEFINITION

Many traditional watch retailers lack a modern digital platform that provides customers with a smooth and convenient shopping experience. Existing platforms are often slow, poorly designed, or lack essential features such as filtering, wish list, and easy checkout.

The challenge is to develop a **responsive, secure, and user-friendly mobile application** that allows users to browse, search, and purchase watches efficiently while maintaining high performance and reliability.

INTRODUCTION

In today's digital era, mobile applications play a vital role in simplifying online shopping experiences. With the rapid growth of e-commerce, customers expect fast, secure, and user-friendly mobile apps to browse and purchase products anytime, anywhere.

Watch Hub is a Flutter-based mobile application designed to provide users with a seamless and modern platform for purchasing premium watches online. The application focuses on usability, performance, and scalability while offering rich features such as product browsing, filtering, cart management, wish list, and secure authentication.

This project has been developed as part of an eProject learning model, allowing students to apply theoretical knowledge into a real-world mobile application using Flutter and Dart.

SCOPE OF PROJECT

The Watch Hub application covers the following scope:

- User registration and login
- Product browsing and searching
- Watch categorization and filtering
- Cart and wishlist functionality
- User profile management
- Reviews and ratings
- Admin product management (conceptual)
- Secure and scalable architecture

BACKGROUND OF THE PROJECT

With the rapid growth of mobile technology and e-commerce, users increasingly prefer mobile applications for shopping due to their convenience, speed, and accessibility. The demand for well-designed and user-friendly shopping applications has increased significantly, especially in the fashion and accessories market.

The WatchHub Mobile Application project was initiated as part of the semester examination to provide us with practical exposure to real-world mobile application development. The idea behind this project is to design and develop a mobile app that allows users to browse, search, and explore a wide range of watches in an organized and interactive manner.

FUNCTIONAL REQUIRMENTS

1 User Authentication

- ◆ Users can register, log in, and log out
- ◆ Secure authentication system
- ◆ Password recovery functionality

2 Browse Watches

- ◆ Browse watches by brand, type, and price
- ◆ Category-wise product listing
- ◆ Sorting by price, popularity, and brand

3 Search and Filters

- ◆ Search watches by name or brand
- ◆ Filters for price range, brand, and type

4 Product Details

- ◆ Detailed product description
- ◆ High-quality product images
- ◆ Price and stock availability

5 Shopping Cart

- ◆ Add watches to cart
- ◆ Update quantity or remove items
- ◆ View total price dynamically

6 Wishlist

- ◆ Save favorite watches
- ◆ Move items from wishlist to cart

7 User Profile

- ◆ View and edit personal details
- ◆ Manage shipping addresses
- ◆ View order history

8 Reviews and Ratings

- ◆ Users can rate products
- ◆ Submit written reviews
- ◆ Sort reviews by date and usefulness

9 Customer Support

- ◆ Contact form for support
- ◆ FAQ section for common queries

10 Admin Panel

- ◆ Manage products
- ◆ View user data
- ◆ Manage orders and reviews

NON-FUNCTIONAL REQUIRMENTS

Performance:

- ◆ App response time should be under 2 seconds

Usability:

- ◆ Simple and intuitive UI

Reliability:

- ◆ Smooth operation without crashes

Scalability:

- ◆ Can handle increased users and products

Security:

- ◆ Secure authentication and user data protection

Accessibility:

- ◆ Clear fonts and easy navigation

Error Handling:

- ◆ User-friendly error messages

Maintainability:

- ◆ Clean and modular code structure

PROPOSED SOLUTION

WatchHub will provide a complete mobile-first e-commerce solution focused on delivering a smooth, secure, and engaging watch shopping experience. The application will be developed using Flutter, enabling cross-platform deployment with a single codebase.

The solution combines modern UI/UX design, efficient state management, and scalable architecture to allow users to browse, search, and purchase watches effortlessly. WatchHub aims to bridge the gap between traditional retail and modern mobile commerce by offering rich product discovery features, personalization, and performance optimization.

WORKFLOW OVERVIEW:

◆ User Entry & Authentication

Users can sign up or log in using secure authentication mechanisms. The system ensures that only registered users can access features such as wishlist, cart, and order history.

◆ Product Discovery

Once authenticated, users can explore a wide catalog of watches categorized by:

- Brand
- Watch type
- Price range

Advanced filtering and sorting options allow users to quickly find watches that match their preferences.

◆ Search & Smart Filtering

Users can search for specific watches using keywords. Filters help narrow results based on brand, price, and popularity, improving product discoverability.

◆ Product Details View

Each watch has a dedicated product page displaying:

- High-quality images
- Detailed specifications
- Price and availability
- User ratings and reviews

This helps users make informed purchase decisions.

◆ Cart & Wishlist Management

Users can:

- Add products to cart
- Modify quantities
- Save items to wishlist for future purchase
- The cart dynamically updates pricing and total cost.

◆ Checkout & Order Tracking (Conceptual)

Users can review cart items and proceed with checkout. Order details are saved and can be viewed later through the user profile section.

◆ User Profile & History

- Each user has a profile where they can:
- Update personal information
- Manage shipping addresses
- View order history

◆ Data Handling & Storage

Application data such as user details, product listings, cart data, and reviews are securely managed through backend services or cloud-based solutions.

DESIGN SPECIFICATION

Frontend (Mobile Application)

Framework & Tools

- Flutter SDK
- Dart Programming Language
- Material Design Components

State Management

Provider / Riverpod / Bloc (conceptual, depending on implementation)

UI/UX Design

- Clean, modern, and minimal design
- Mobile-first responsive layouts
- Smooth navigation with bottom navigation bar
- Consistent color scheme aligned with premium watch branding
- Reusable components for cards, buttons, and lists

Core Screens Include:

- Splash Screen
- Login / Signup
- Home (Product Listing)
- Product Details
- Search & Filters
- Cart
- Wishlist
- Profile

Accessibility

- Clear typography and readable fonts
- Touch-friendly UI elements
- Consistent navigation patterns
- Error messages with user-friendly feedback

Backend

1 Framework

Supabase / REST APIs (conceptual)

2 Core Services

- Authentication Service
- Product Management Service
- Order & Cart Service
- Review & Rating Service

3 Database Design

- User data storage
- Product catalog storage
- Cart and wishlist data

- Reviews and ratings
- Data is structured to ensure scalability and quick retrieval.

4 Security Measures

- Secure authentication
- Data validation
- Restricted access to user-specific data
- Secure storage of sensitive information

5 Key Features

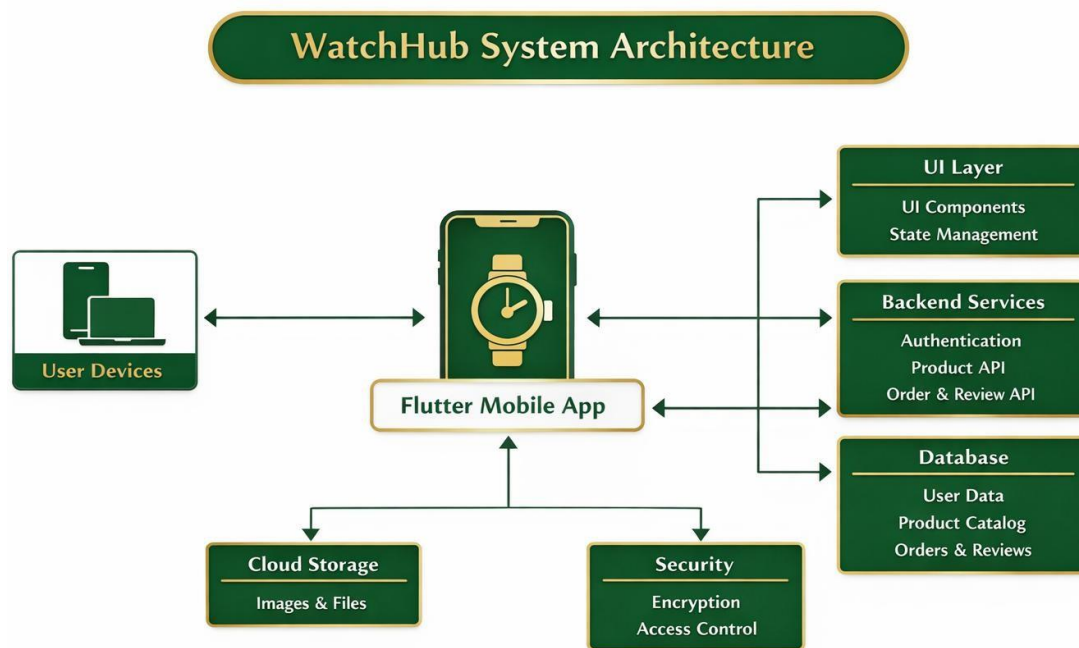
- User authentication & profile management
- Product browsing with filters and search
- Wishlist and cart functionality
- Reviews and ratings
- Responsive and intuitive UI
- Scalable and maintainable architecture

6 History & Data Persistence

- User cart and wishlist persistence
- Order history stored for future reference
- Profile data saved securely

7 Future Enhancements

- Online payment gateway integration
- Push notifications for offers and updates
- Admin dashboard for inventory management
- AI-based product recommendations
- Multi-language support



DETAIL STEPS TO EXECUTE THE PROJECT

This section describes the step-by-step process followed to design, develop, and deploy the WatchHub Mobile Application using Flutter.

TECHNOLOGY REQUIREMENTS

Backend Requirements

Install Supabase SDK or REST API backend

Authentication Service (Supabase Auth / Custom API)

Database:

- Cloud Firestore / MongoDB / MySQL (conceptual)

Storage Service:

- Supabase Storage / Cloud Storage for product images

Environment Configuration:

- API keys
- Database connection strings

Frontend (Mobile App) Requirements

Install Flutter SDK (latest stable)

Install Dart SDK

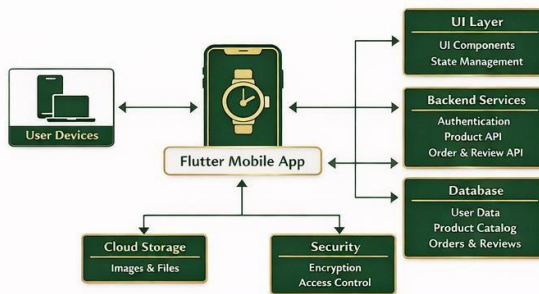
IDE:

➤ Android Studio / VS Code

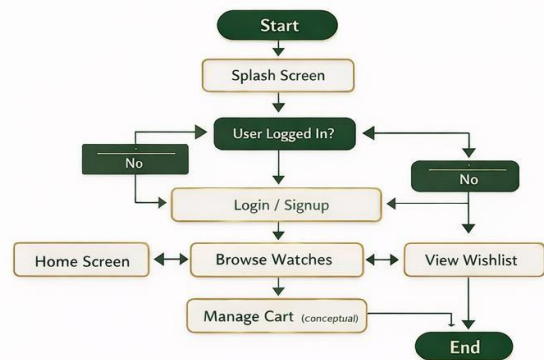
Android Emulator or Physical Android Device

DIAGRAM

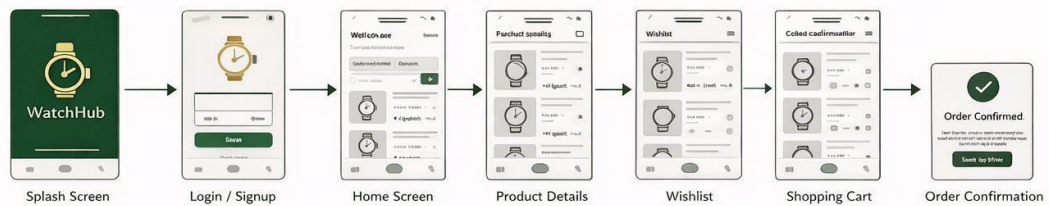
WatchHub System Architecture



User Flow - WatchHub Mobile Application

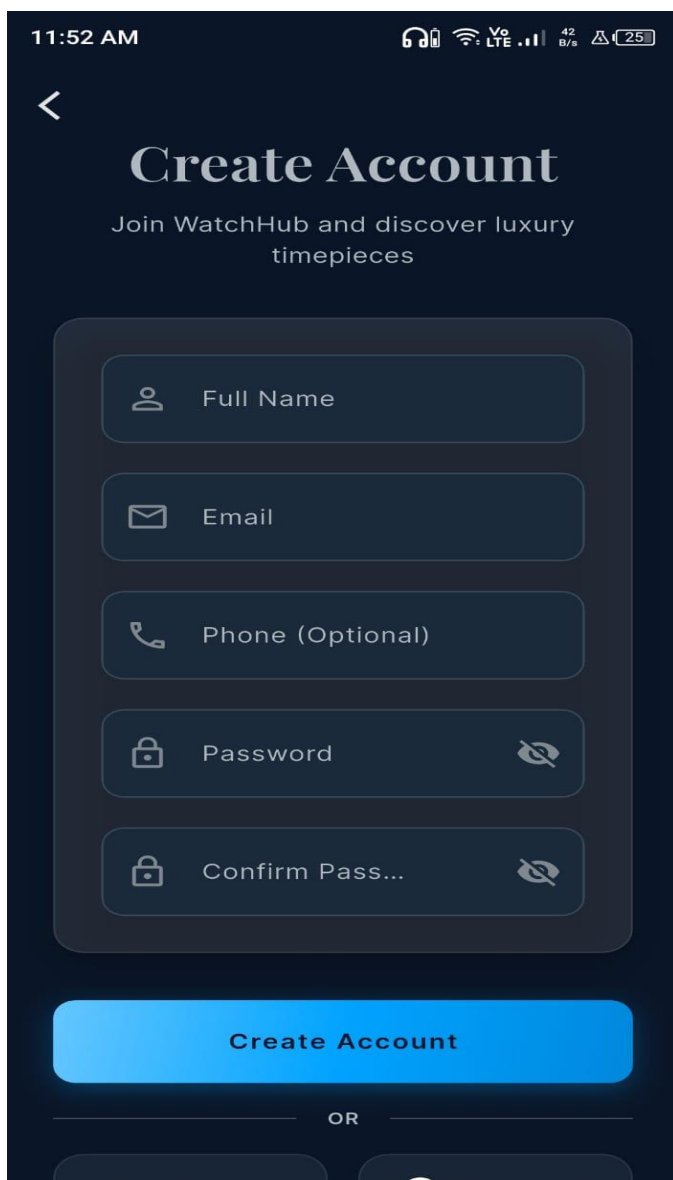


WatchHub - UI Wireframe



SCREENSHOTS

1. The app shall display a Splash Screen that checks the user's login status and redirects accordingly.



11:52 AM

<

Create Account

Join WatchHub and discover luxury timepieces

Full Name

Email

Phone (Optional)

Password

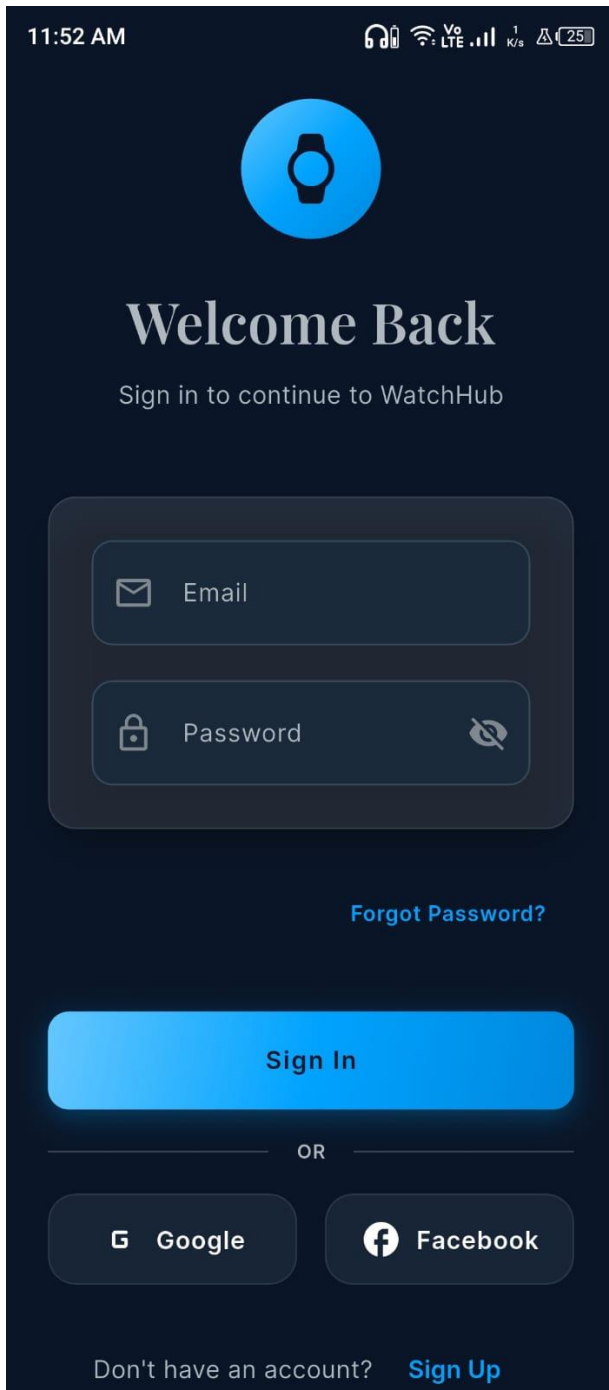
Confirm Pass...

Create Account

OR

User Authentication:

- a. Users must be able to create accounts, log in, and log out.
- b. Authentication should be secure and include password recovery options.



11:52 AM

🔊 📶 VoLTE .11 1 K/s 📶 25



Welcome Back

Sign in to continue to WatchHub

 Email

 Password 

[Forgot Password?](#)

Sign In

OR

 Google  Facebook

Don't have an account? [Sign Up](#)

11:52 AM   VoLTE  42 B/s  25

<

 Full Name

 Email

 Phone (Optional)

 Password 

 Confirm Pass... 

Create Account

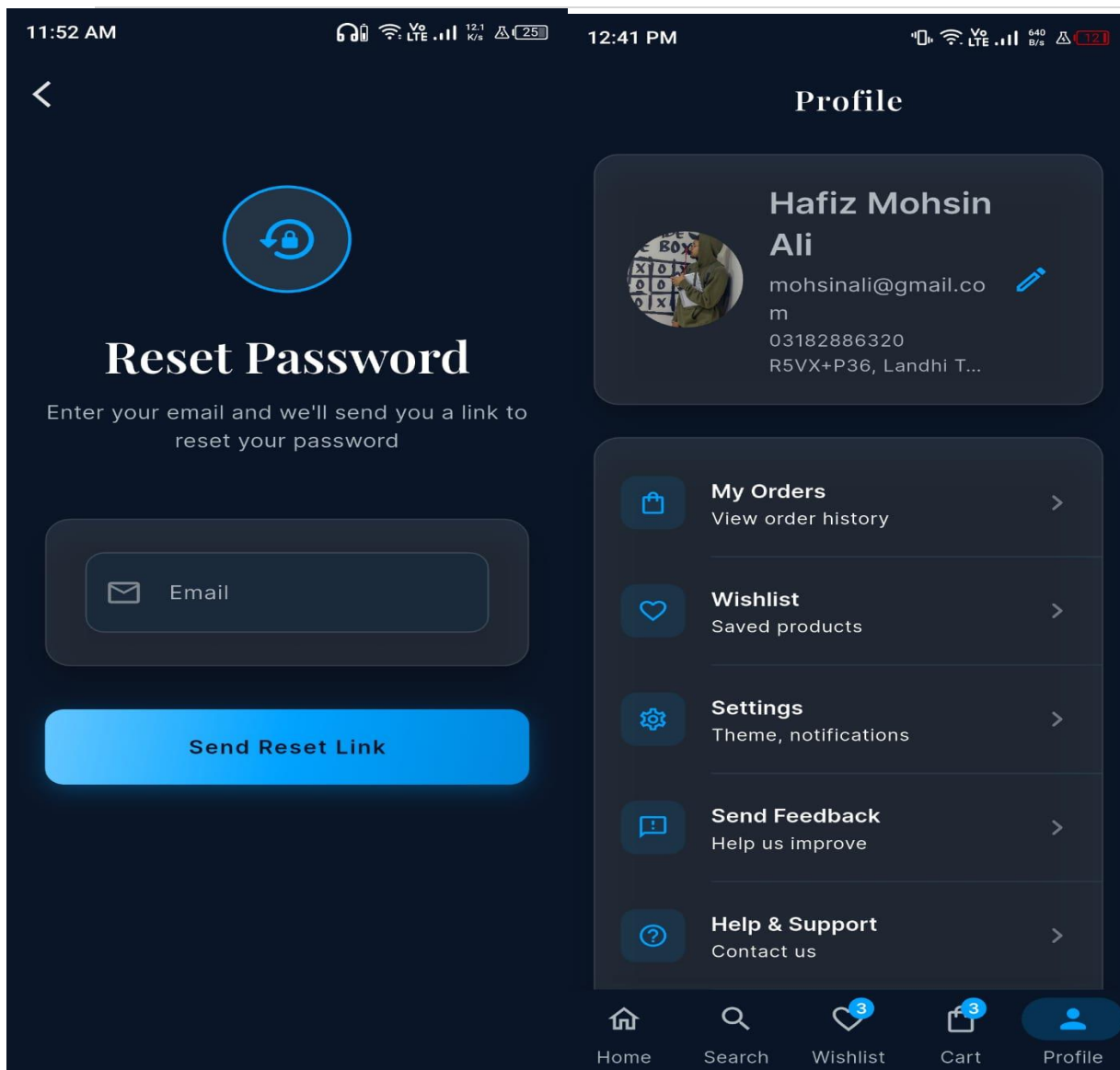
OR

 Google  Facebook

Already have an account? [Sign In](#)

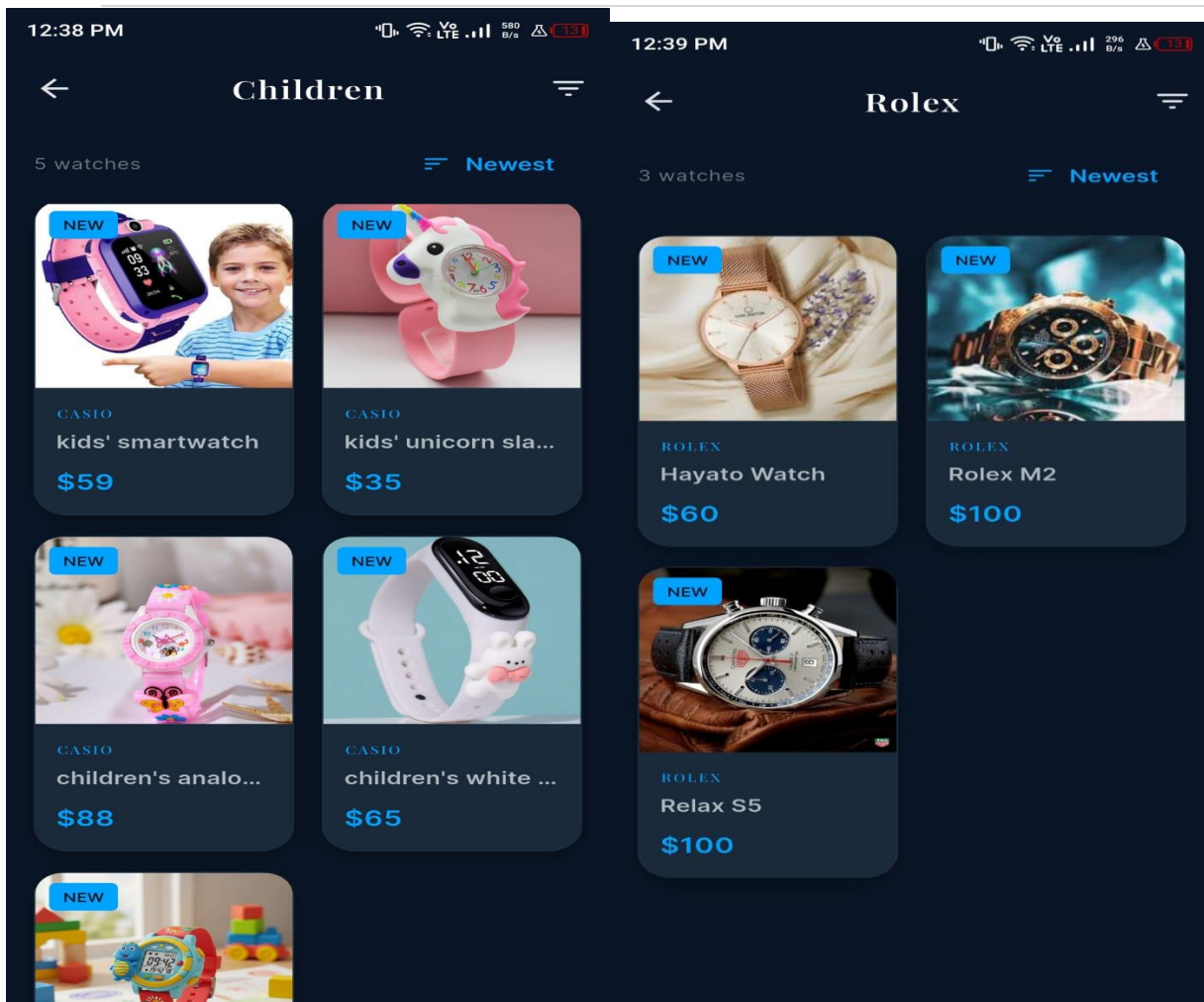
User Profiles:

- Users can create and edit their profiles, including personal information, and shipping addresses.
- Access to order history and order tracking.

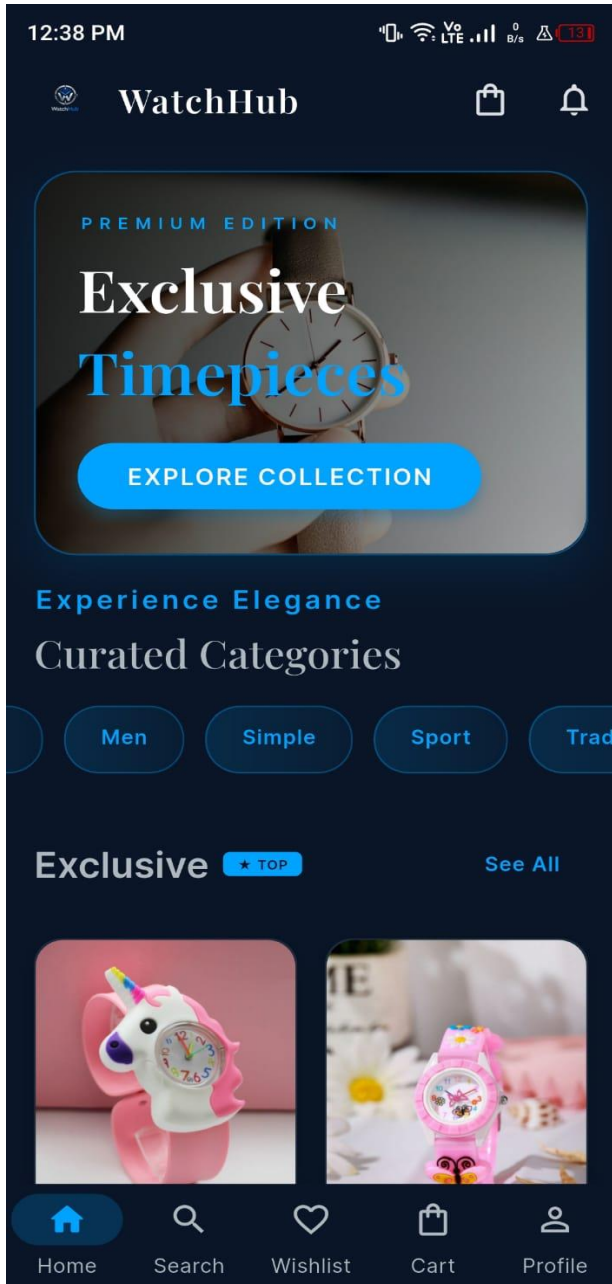


Browse Watches:

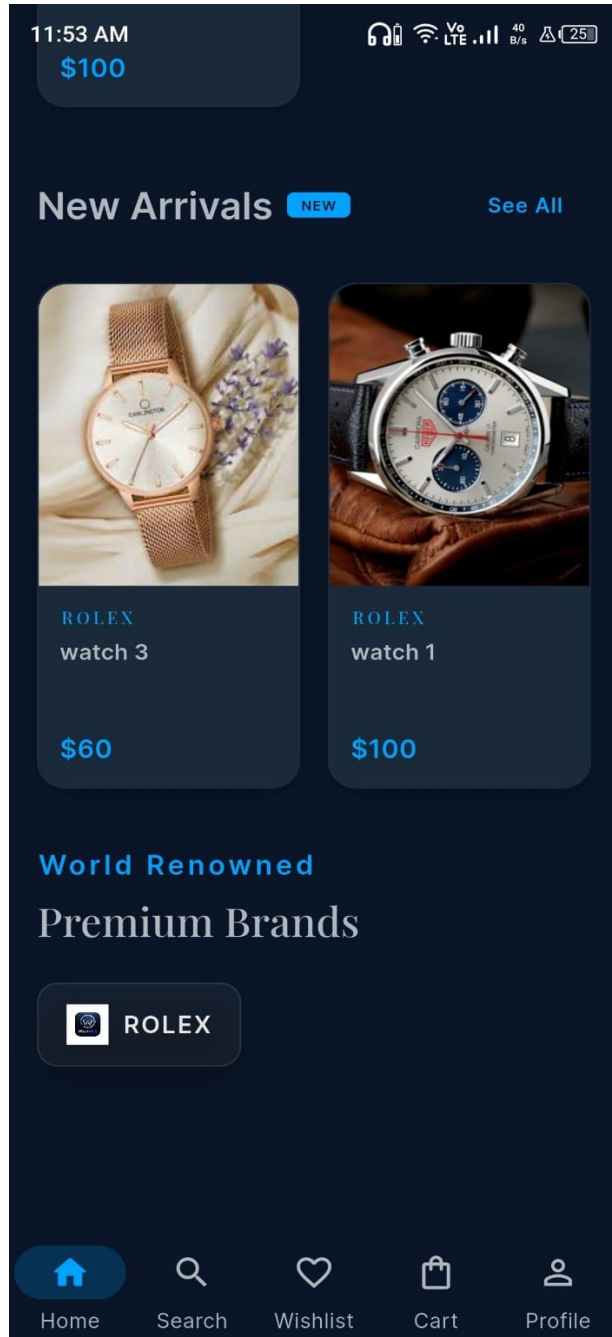
- Users can browse a catalog of watches, categorized by brands, types, and price ranges.
- Filtering options should be available, including sorting by price, brand, and popularity.



PREMIUM EDITION



NEW ARRIVALS



RESET PASWORD

11:52 AM







Reset Password

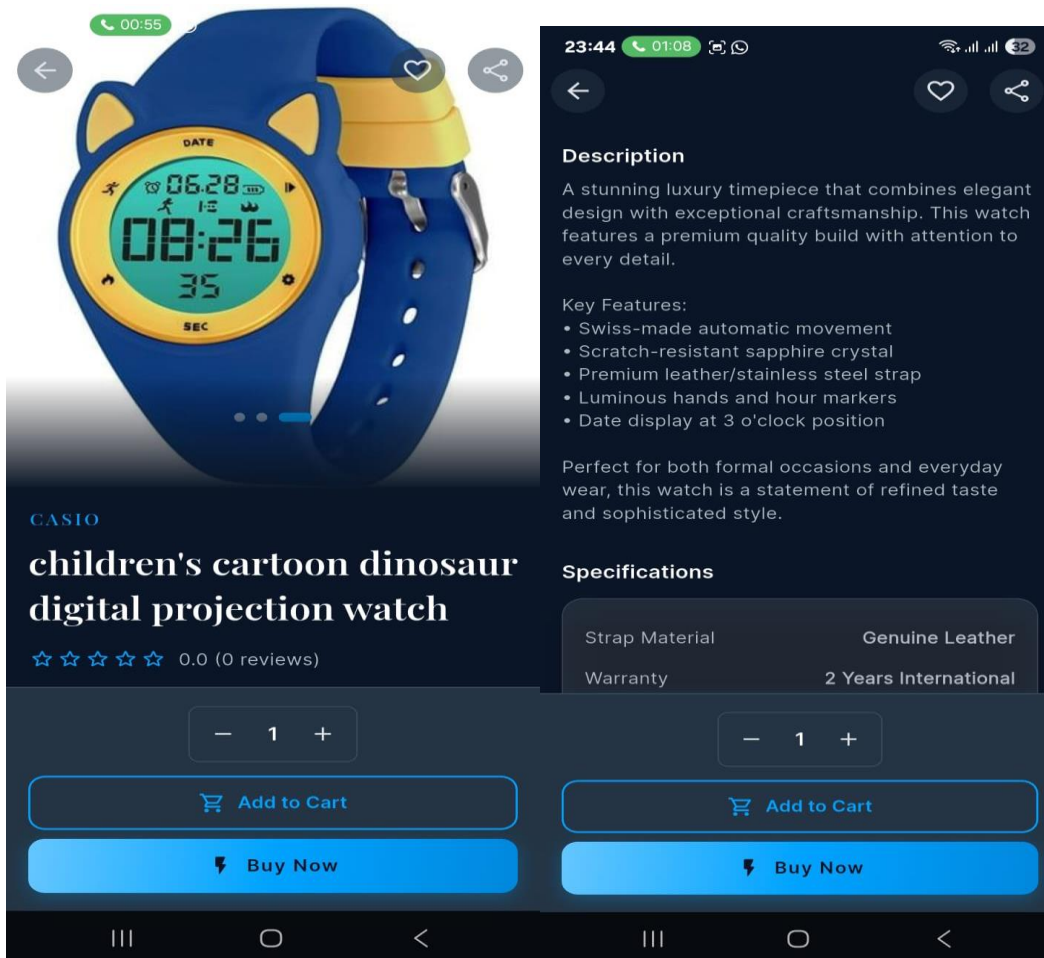
Enter your email and we'll send you a link to reset your password

 Email

Send Reset Link

Product Details:

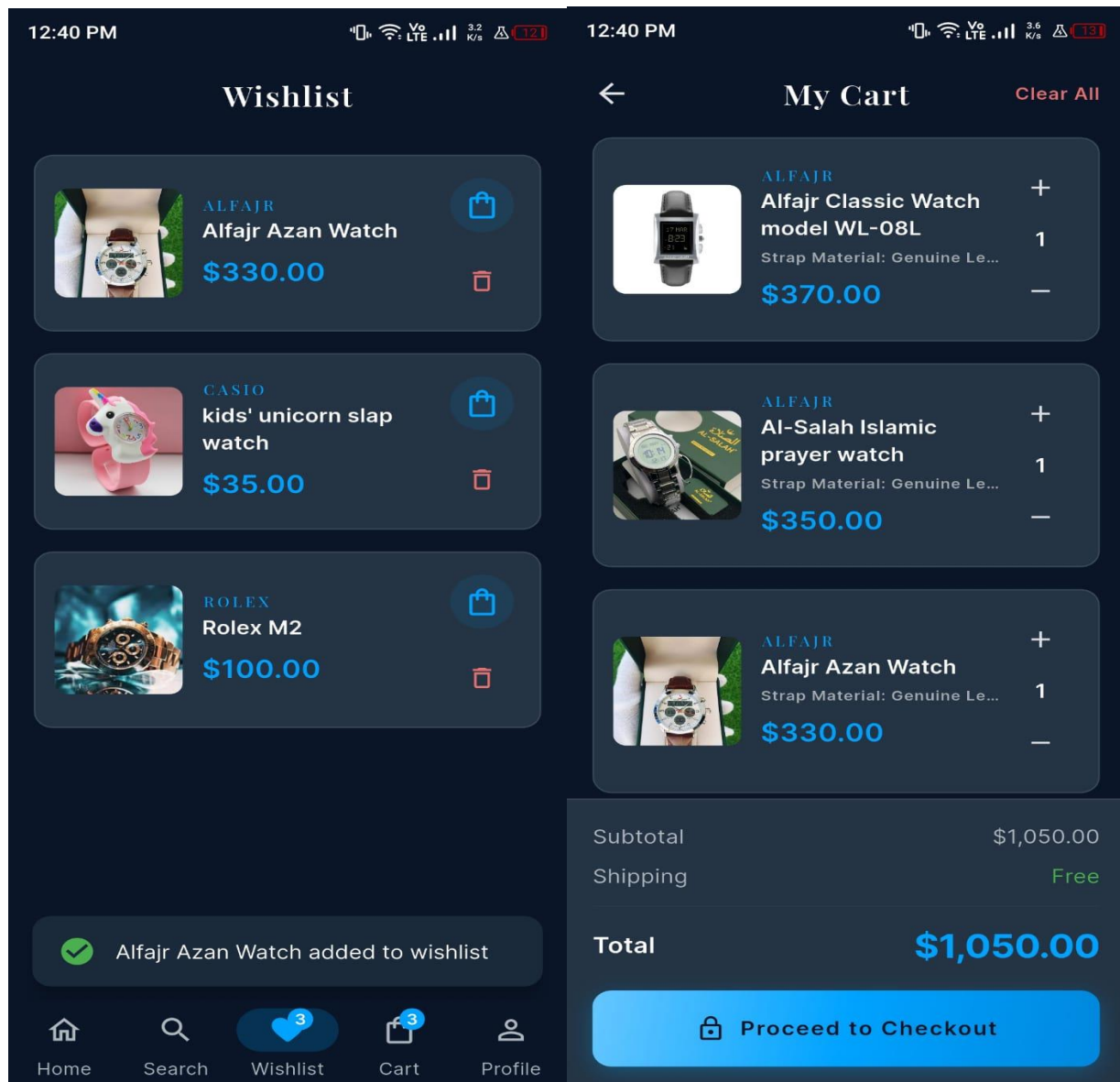
- Detailed product pages for each watch, including images, descriptions, specifications, and prices.
- Users can view high-quality images of the watches and zoom in for a closer look.
- Availability and stock information should be displayed.





Shopping Cart:

- Users can add watches to their cart.
- Ability to modify the cart contents (add, remove, change quantities).
- Cart should display the total price.



My Orders:

12:42 PM

VoLTE

399 B/s

121

←

My Orders

WH-2026-000021

PENDING

Date

January 30, 2026

Items

3 items

Total

\$1,050.00

Order Details:

12:42 PM

←

Order Details

WH-2026-000021

PENDING

Placed on January 30, 2026

Items

Alfajr Classic Watch model WL-08L

\$370.00

Qty: 1

Al-Salah Islamic prayer watch

\$350.00

Qty: 1

Alfajr Azan Watch

\$330.00

Qty: 1

Shipping Address

Hafiz Mohsin Ali

Hafiz Mohsin Ali

R5VX+P36, Landhi Town, Karachi,

Pakistan

karachi, N/A 273638

USA

Order Summary

Subtotal

\$1,050.00

Shipping

\$0.00

Tax

\$0.00

Total

\$1,050.00

⊗ Cancel Order

12:42 PM

←

Order Details

Al-Salah Islamic prayer watch

\$350.00

Qty: 1

Alfajr Azan Watch

\$330.00

Qty: 1

Shipping Address

Hafiz Mohsin Ali

Hafiz Mohsin Ali

R5VX+P36, Landhi Town, Karachi,

Pakistan

karachi, N/A 273638

USA

Order Summary

Subtotal

\$1,050.00

Shipping

\$0.00

Tax

\$0.00

Total

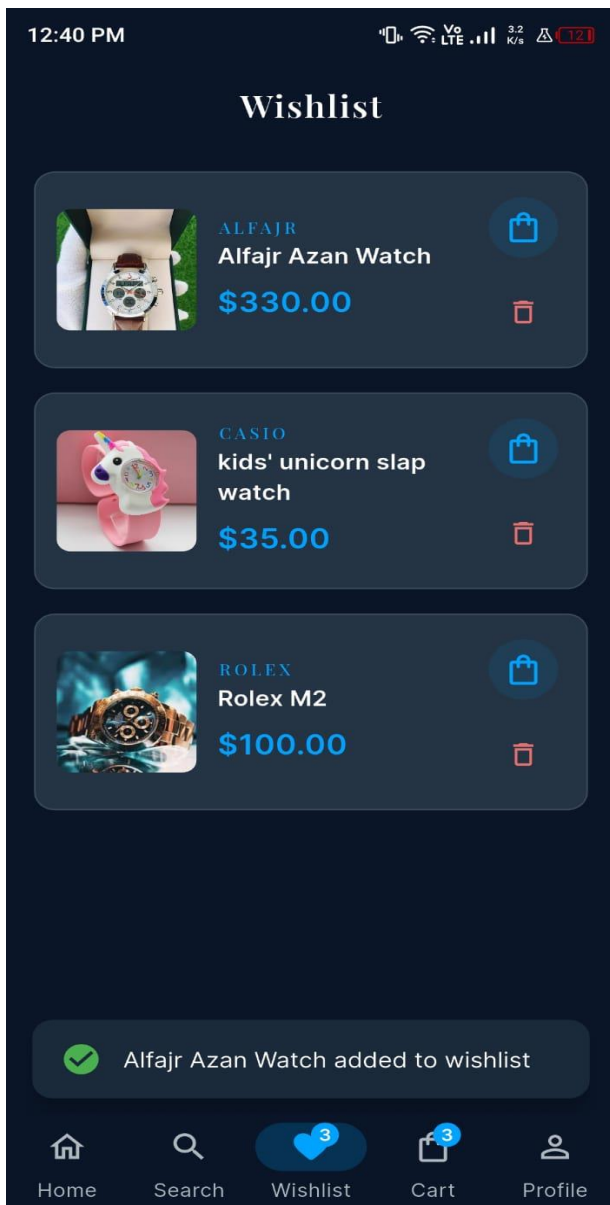
\$1,050.00

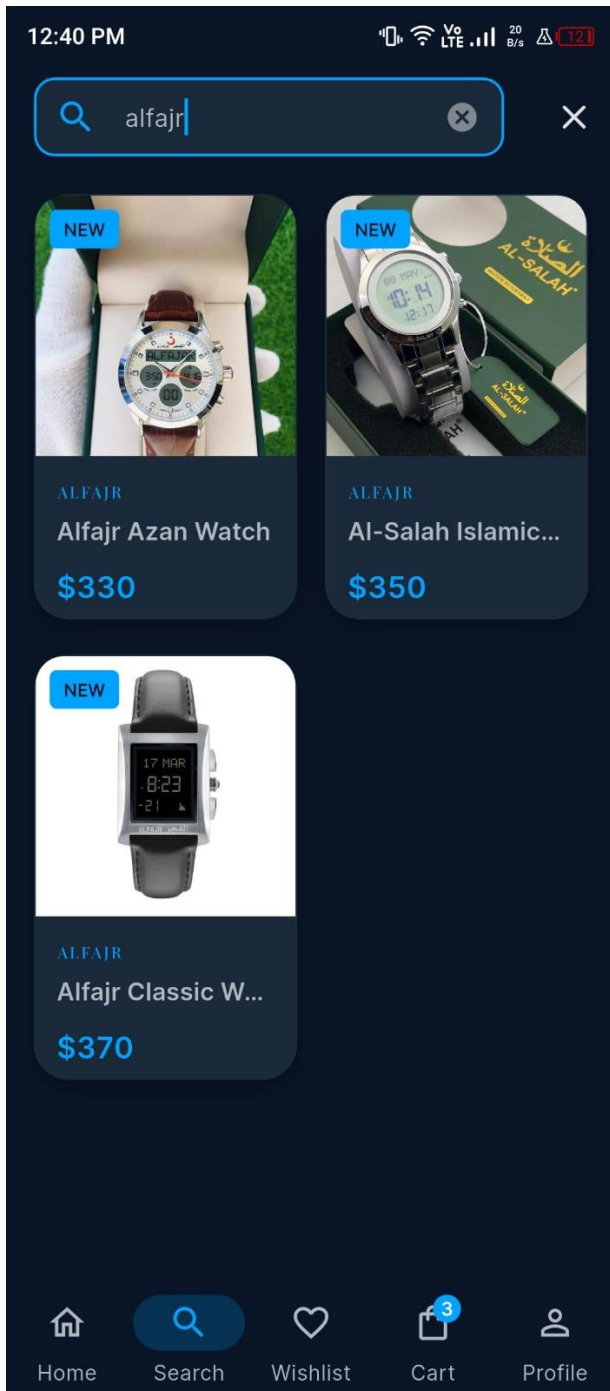
⊗ Cancel Order



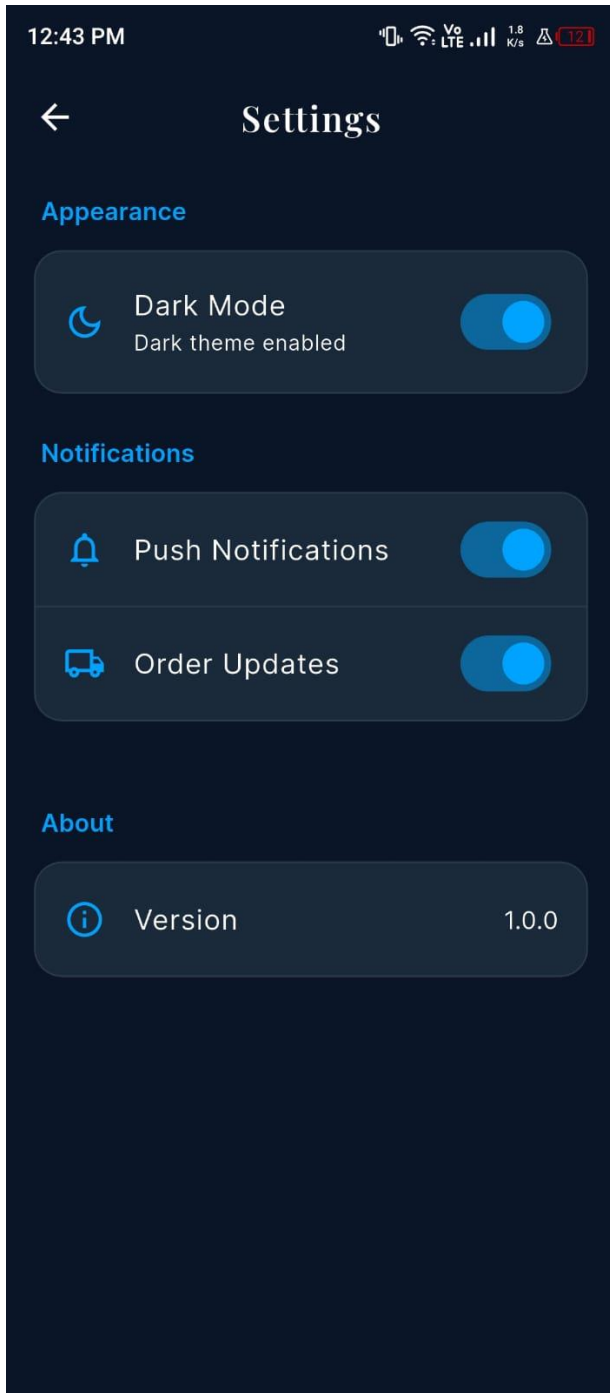
Wishlist:

- Users can add watches to a wishlist for future reference.
- Wishlist items can be moved to the cart for purchase.



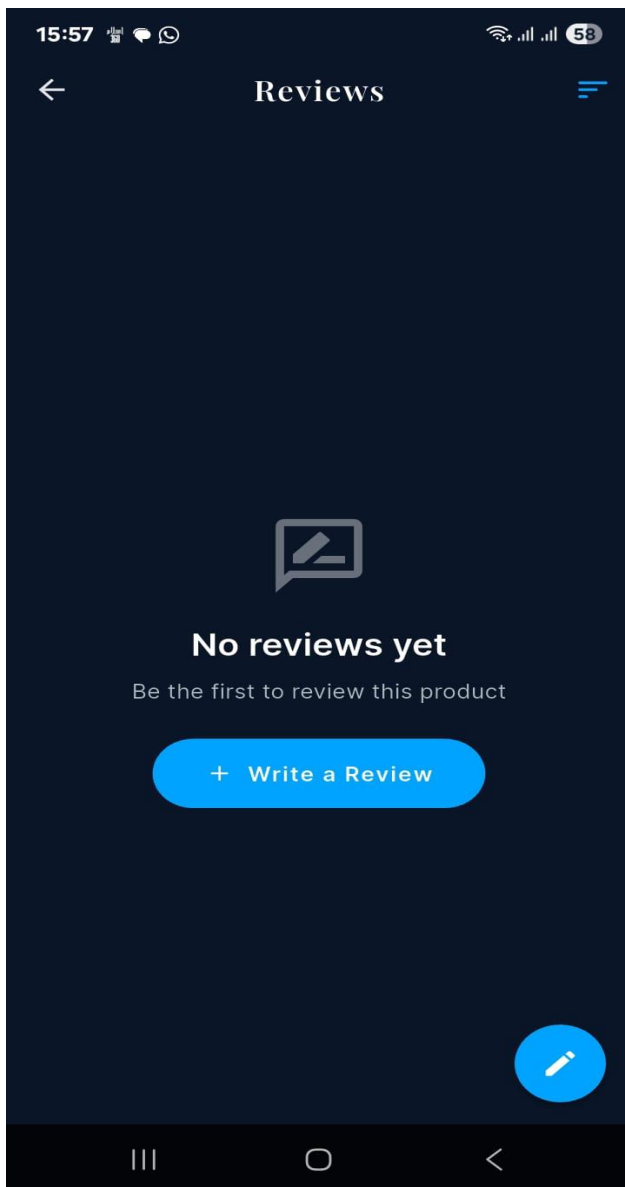


SETTINGS



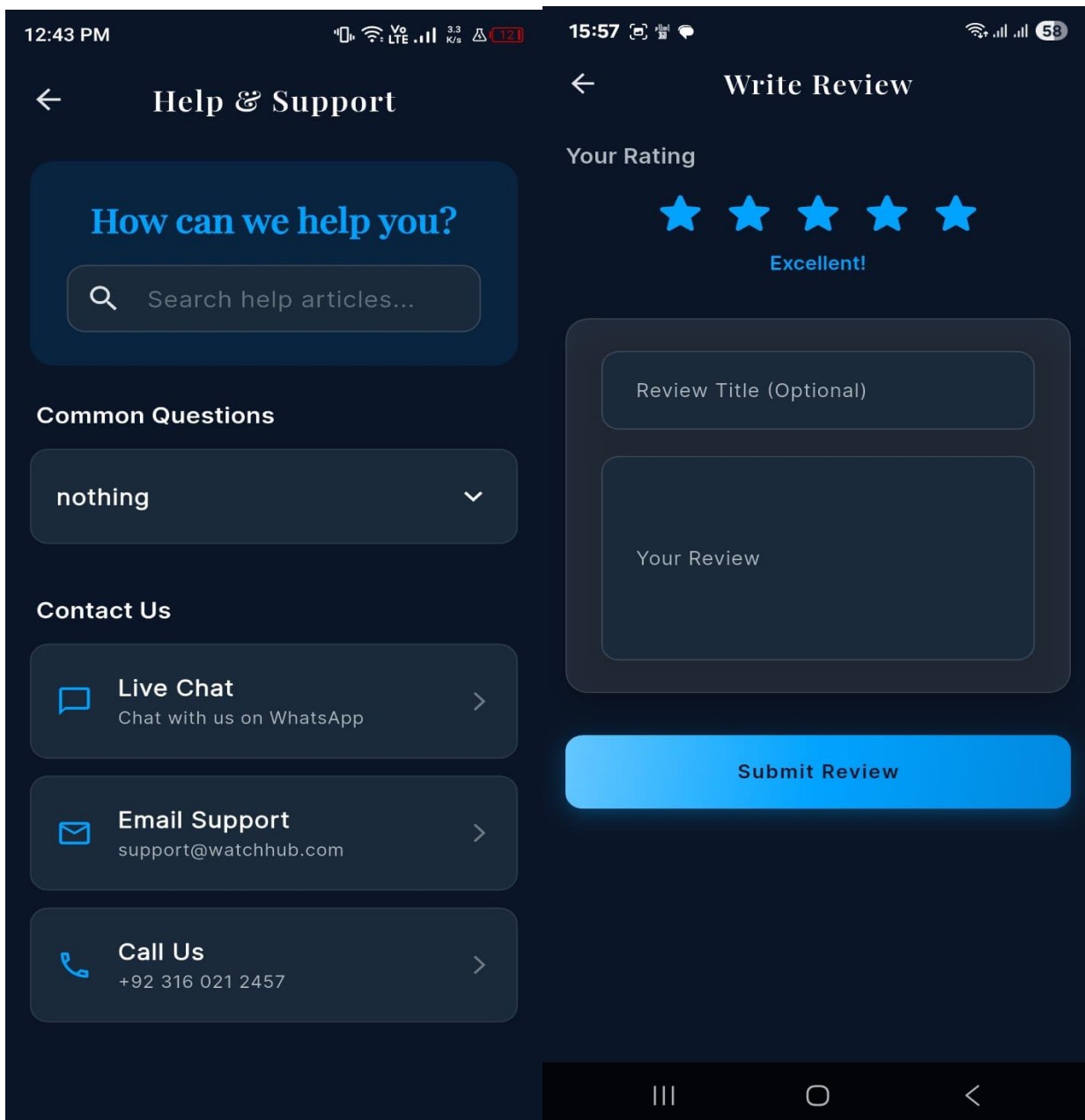
Reviews and Ratings:

- Users can leave reviews and ratings for watches.
- Sort and filter reviews based on helpfulness and date.



Customer Support:

- Contact customer support through an in-app chat or a contact form.
- View a frequently asked questions (FAQ) section.



Help & Support

How can we help you?

Search help articles...

Common Questions

nothing

Contact Us

Live Chat
Chat with us on WhatsApp

Email Support
support@watchhub.com

Call Us
+92 316 021 2457

Write Review

Your Rating

★★★★★

Excellent!

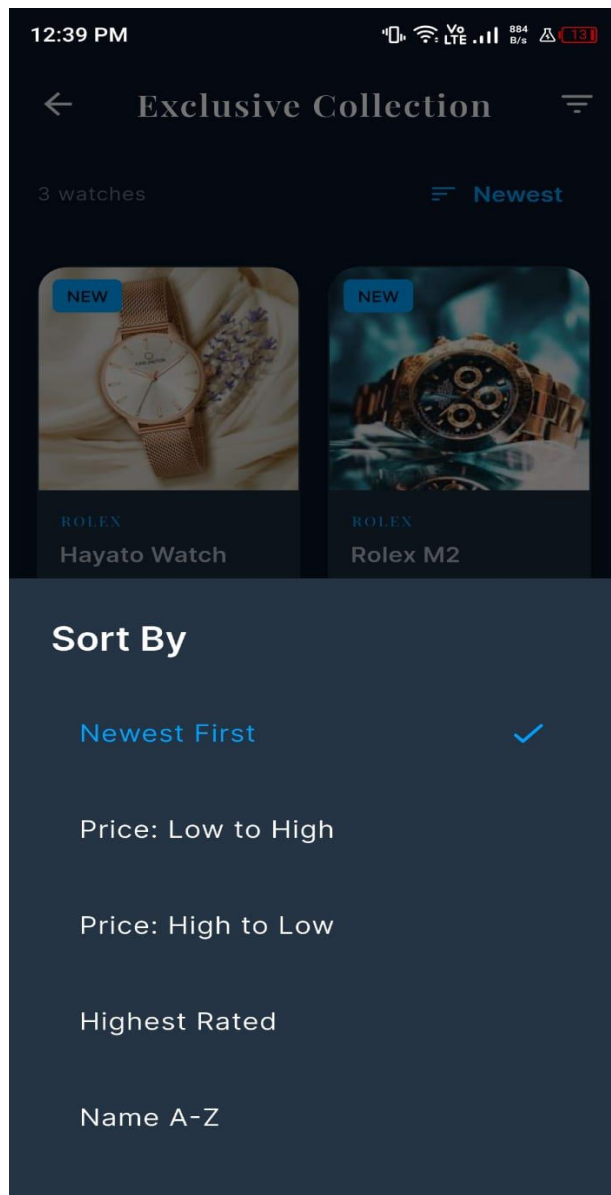
Review Title (Optional)

Your Review

Submit Review

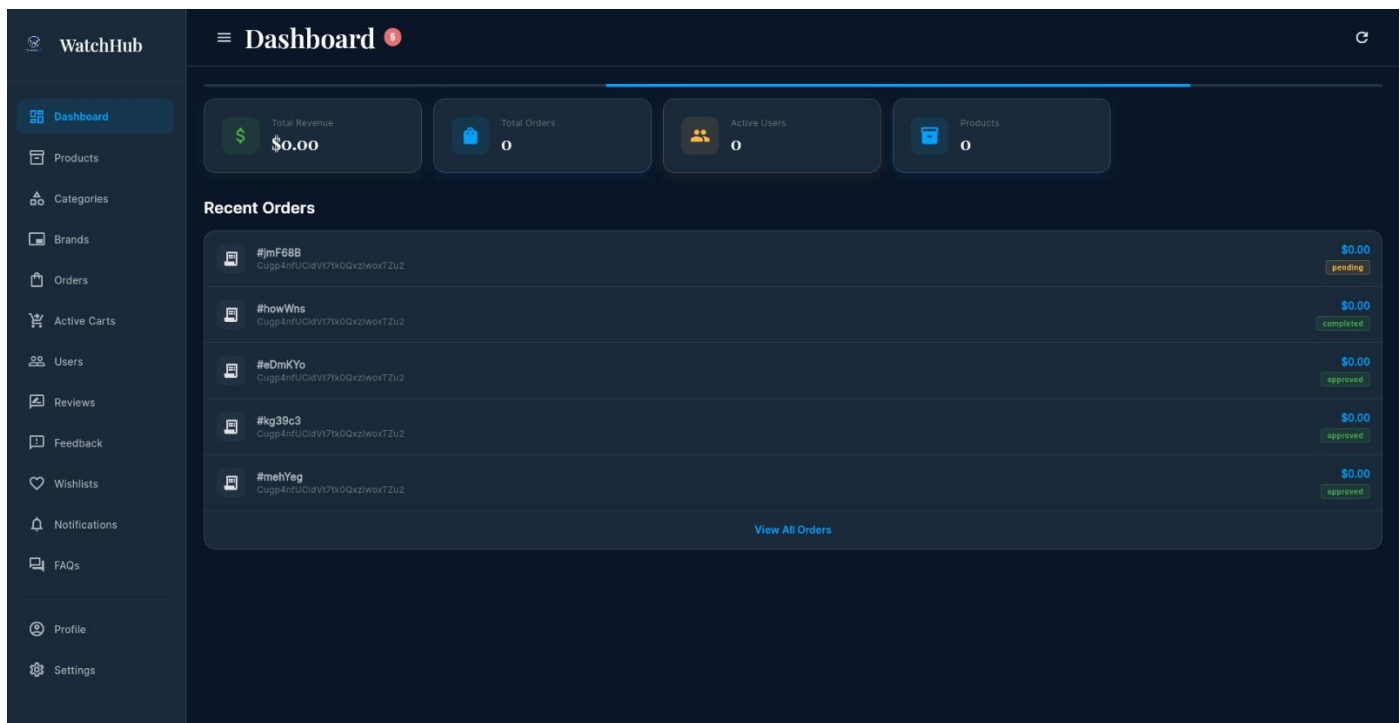
Search and Filters:

- Robust search functionality.
- Filters for narrowing down search results, including brand, price, and product type.



Admin Panel:

A backend admin panel to manage product listings, user data, reviews, and orders.



The screenshot displays the WatchHub Admin Dashboard. The left sidebar contains a navigation menu with the following items: Dashboard, Products, Categories, Brands, Orders, Active Carts, Users, Reviews, Feedback, Wishlists, Notifications, FAQs, Profile, and Settings. The main content area is titled 'Dashboard' and features four summary cards: Total Revenue (\$0.00), Total Orders (0), Active Users (0), and Products (0). Below these cards is a 'Recent Orders' section with a table of five orders. Each order row includes an ID, a user identifier, a status badge, and a price of \$0.00. A 'View All Orders' link is positioned at the bottom of the table.

ID	User	Status	Price
#jmf88B	Cugg4nFUcIdV17k0Qx2lweXTZu2	pending	\$0.00
#howWns	Cugg4nFUcIdV17k0Qx2lweXTZu2	completed	\$0.00
#eDmkYo	Cugg4nFUcIdV17k0Qx2lweXTZu2	approved	\$0.00
#kg39c3	Cugg4nFUcIdV17k0Qx2lweXTZu2	approved	\$0.00
#mehYeg	Cugg4nFUcIdV17k0Qx2lweXTZu2	approved	\$0.00

WatchHub

Dashboard

Products

Categories

Brands

Orders

Active Carts

Users

Reviews

Feedback

Wishlists

Notifications

FAQs

Profile

Settings

Categories

Children

Classic

Luxury

Men

Simple

Sport

Traditional

Women

WatchHub

Products

Image	Name	Brand	Price	Stock	Category	Actions
	watch 3	Rolex	\$60	0	Children	Edit Add Delete
	Watch 2	Rolex	\$100	0	Sport	Edit Add Delete
	watch 1	Rolex	\$100	0	Simple	Edit Add Delete

WatchHub

Dashboard

Products

Categories

Brands

Orders

Active Carts

Users

Reviews

Feedback

Wishlists

Notifications

FAQs

Profile

Settings

≡ Orders 5

Order ID	Date	Customer	Total	Status	Items	Actions
#jmf68B	01/29/26	Cugp4nfUCldVl7tk0QxzIwoxTZu2	\$240.00	PENDING	1 items	Pending
#howWns	01/29/26	Cugp4nfUCldVl7tk0QxzIwoxTZu2	\$60.00	COMPLETED	1 items	Completed
#eDmKYe	01/29/26	Cugp4nfUCldVl7tk0QxzIwoxTZu2	\$100.00	APPROVED	1 items	APPROVED
#kg39c3	01/29/26	Cugp4nfUCldVl7tk0QxzIwoxTZu2	\$100.00	APPROVED	1 items	APPROVED
#mehYeg	01/29/26	Cugp4nfUCldVl7tk0QxzIwoxTZu2	\$500.00	APPROVED	1 items	APPROVED
#WIDzmK	01/28/26	YDhumx33RBYXGZRwBPPUdcFxC42	\$250.00	PENDING	1 items	Pending
#xYZ5Wb	01/28/26	YDhumx33RBYXGZRwBPPUdcFxC42	\$300.00	APPROVED	1 items	APPROVED
#OZU4sR	01/27/26	YDhumx33RBYXGZRwBPPUdcFxC42	\$3750.00	APPROVED	1 items	APPROVED
#wNoh8S	01/27/26	YDhumx33RBYXGZRwBPPUdcFxC42	\$72750.00	CANCELLED	1 items	Cancelled
#5CZ0EV	01/27/26	YDhumx33RBYXGZRwBPPUdcFxC42	\$131850.00	PENDING	1 items	Pending
#fxt74U	01/27/26	YDhumx33RBYXGZRwBPPUdcFxC42	\$250000.00	PENDING	1 items	Pending
#ptVfKy	01/27/26	h5TMEfZUCLSh8qsQiPr7VR9jeQ92	\$8200.00	PENDING	1 items	Pending
#iI2pk	01/26/26	FSZ7PkTghrWjYToFIeeHzSolDV2	\$52950.00	PENDING	1 items	Pending
#mlhzoD	01/26/26	KuzzVsZaibhZLyIcBpgRTJfvJUJ3	\$125000.00	APPROVED	1 items	APPROVED
#kgDIZe	01/24/26	FSZ7PkTghrWjYToFIeeHzSolDV2	\$14550.00	APPROVED	1 items	APPROVED
#OObYJz	01/24/26	jWO7M7xda2RH1izXDLh9Mbhhkp1	\$5400.00	PENDING	1 items	Pending
#SkrX3o	01/24/26	FSZ7PkTghrWjYToFIeeHzSolDV2	\$12700.00	APPROVED	1 items	APPROVED

WatchHub

Dashboard

Products

Categories

Brands

Orders

Active Carts

Users

Reviews

Feedback

Wishlists

Notifications

FAQs

Profile

Settings

≡ Manage Brands 5

ROLEX

WatchHub

Dashboard

Products

Categories

Brands

Orders

Active Carts

Users

Reviews

Feedback

Wishlists

Notifications

FAQs

Profile

Settings

Users

5

Name

Email

Role

Joined

Actions

A

y@gmail.com

CUSTOMER

Jan 27, 2026

Miss Kiran

kiran@gmail.com

CUSTOMER

Jan 27, 2026

Mohsin Ali

mohsin@gmail.com

CUSTOMER

Jan 27, 2026

Affan

b@gmail.com

CUSTOMER

Jan 26, 2026

zain

zain1234@gmail.com

CUSTOMER

Jan 24, 2026

Aqsa Ahmed

aqsaaptch19@gmail.com

CUSTOMER

Jan 24, 2026

Contactnexusvate

contactnexusvate@gmail.com

CUSTOMER

Jan 23, 2026

Muhammad Affan

affan.work05@gmail.com

CUSTOMER

Jan 23, 2026

Affan

m@gmail.com

CUSTOMER

Jan 22, 2026

Affan

muhammadaffan1445@gmail.com

CUSTOMER

Jan 02, 2026

Hafiz Mohsin Ali

hafizmohsin.at@gmail.com

CUSTOMER

Dec 29, 2025

Affan

a@gmail.com

ADMIN

Dec 29, 2025

Super Admin

admin@watchhub.com

ADMIN

Dec 29, 2025

Super Admin

Affan

affan@gmail.com

CUSTOMER

Dec 29, 2025

WatchHub

Dashboard

Products

Categories

Brands

Orders

Active Carts

Users

Reviews

Feedback

Wishlists

Notifications

FAQs

Profile

Settings

Active Carts

5

Hafiz Mohsin Ali (hafizmohsin.at@gmail.com)

1 items in cart

Contactnexusvate (contactnexusvate@gmail.com)

1 items in cart

Affan (affan@gmail.com)

1 items in cart

WatchHub

Dashboard

Products

Categories

Brands

Orders

Active Carts

Users

Reviews

Feedback

Wishlists

Notifications

FAQs

Profile

Settings

Feedback

muhammaddaffan1445@gmail.com

affan@gmail.com

WatchHub

Dashboard

Products

Categories

Brands

Orders

Active Carts

Users

Reviews

Feedback

Wishlists

Notifications

FAQs

Profile

Settings

Reviews

Unknown Product

by Affan

abcdefghijkl

Jan 27, 2026

★★★★★

Delete

Reply

Unknown Product

by Mohsin Ali

Nice Quality

Jan 27, 2026

★★★★★

Delete

Reply

Unknown Product

by Affan

nothingnnnn

Jan 28, 2026

★★★★★

Admin Reply

thanks for the reviews

Delete

Edit Reply

Unknown Product

by Aqsa Ahmed

abccogqsohwxhlwxiwhdltsehsusu

Jan 24, 2026

★★★★★

WatchHub

Dashboard

Products

Categories

Brands

Orders

Active Carts

Users

Reviews

Feedback

Wishlists

Notifications

FAQs

Profile

Settings

Watch 2

ROLEX

Affan

OUT OF STOCK

Jan 29, 2026

watch 1

ROLEX

Affan

OUT OF STOCK

Jan 29, 2026

Unknown Product

Mohsin Ali

OUT OF STOCK

Jan 27, 2026

Unknown Product

Mohsin Ali

OUT OF STOCK

Jan 27, 2026

Unknown Product

Mohsin Ali

OUT OF STOCK

Jan 27, 2026

Unknown Product

Mohsin Ali

OUT OF STOCK

Jan 27, 2026

Unknown Product

Contactnexusvate

OUT OF STOCK

Jan 24, 2026

New Order: WH-2026-000020

Affan placed an order for \$240.00

Jan 29, 2026 19:31

New Wishlist Item

Muhammad Affan added "Watch 2" to their wishlist

Jan 29, 2026 19:30

New Order: WH-2026-000019

Affan placed an order for \$80.00

Jan 29, 2026 12:45

New Order: WH-2026-000018

Affan placed an order for \$100.00

Jan 29, 2026 12:35

New Order: WH-2026-000017

Affan placed an order for \$100.00

Jan 29, 2026 12:27

New Order: WH-2026-000016

Affan placed an order for \$500.00

Jan 29, 2026 12:19

New Wishlist Item

47 | Page

The image displays a web application interface with two main sections. The top section, titled "FAQ Management", features a sidebar on the left with navigation links: Dashboard, Products, Categories, Brands, Orders, Active Carts, Users, Reviews, Feedback, Wishlists, Notifications, FAQs, Profile, and Settings. The main content area shows a list of four FAQ entries, each with a title and a description, and edit/delete icons. The bottom section, titled "Profile", shows a user profile for "Super Admin" with the email "admin@watchhub.com". It includes a circular profile picture placeholder and three buttons: "Edit Profile", "Reset Password", and "Sign Out".

WatchHub

Dashboard

Products

Categories

Brands

Orders

Active Carts

Users

Reviews

Feedback

Wishlists

Notifications

FAQs

Profile

Settings

Settings

APPEARANCE

APPEARANCE

Dark Mode
Toggle dark/light theme

ACCOUNT

Email
admin@watchhub.com

Role
Super Admin

SESSION

Sign Out

WatchHub Admin v1.0.0

WatchHub

Dashboard

Total Revenue

\$1053850.00

Total Orders

20

Active Users

15

Products

3

Categories

8

Brands

1

Reviews

0

Feedback

0

Sales Analytics

Recent Orders

#jmF68B

Cu9p4nFUChdV17kOGxzIwoxTZu2

\$0.00

pending

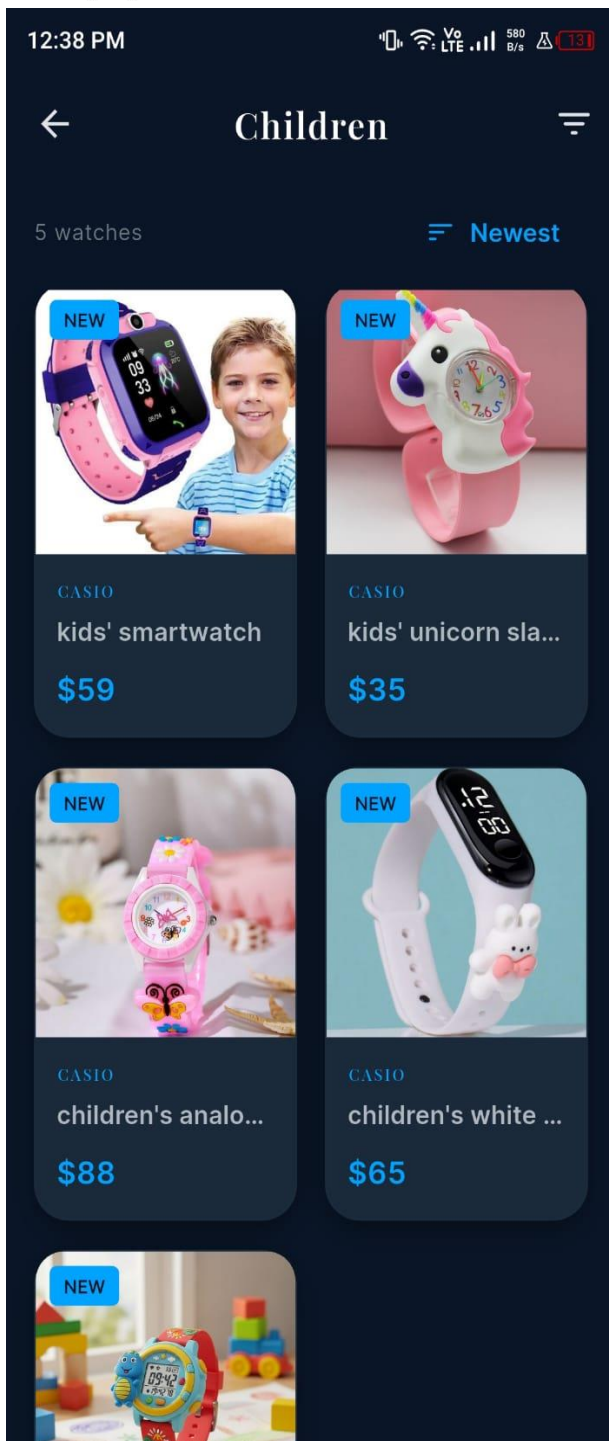
#howWns

Cu9p4nFUChdV17kOGxzIwoxTZu2

\$0.00

completed

Childern Section:



Feedback and Reviews:

Users can provide feedback and report issues directly from the app.

12:43 PM

← **Send Feedback**

We value your feedback!

Help us improve WatchHub by sharing your thoughts, reporting issues, or suggesting new features.

Feedback Type

General Feedback

Report a Bug

Feature Request

Complaint

... Other

Subject

Message

Submit Feedback

HARDWARE REQUIREMENTS

Hardware:

- Pentium IV or higher
- Minimum 4 GB RAM
- 10 GB free disk space
- Internet connection

SOFTWARE REQUIREMENTS

Software:

- Windows 10 or higher
- Flutter SDK
- Dart SDK
- Android Studio / VS Code
- Android Emulator